TRƯỜNG ĐẠI HỌC FPT

FPT UNIVERSITY · DEPARTMENT OF INFORMATION SYSTEMS
COURSE: KMS301 — KNOWLEDGE MANAGEMENT SYSTEMS
ACADEMIC TERM: FALL 2026 · INSTRUCTOR: INFORMATION SYSTEMS FACULTY

ASSIGNMENT 1: STRATEGIC KNOWLEDGE MANAGEMENT

STRATEGIC KM ARCHITECTURE & DOMAIN DECOMPOSITION REPORT

Project: Domain-Specific Knowledge Management System for I-Ching Heritage (KinhDich_KB)
From Fragmented Folklore to an Auditable Corporate Knowledge Asset

Target Organization (Target Domain)	The Vietnamese I-Ching Scholar & Practitioner Community
Academic Unit	Department of Information Systems — FPT University Hanoi
Course & Class	KMS301 (Knowledge Management Systems) · Class IS1906
Project Team	Team 05 (06 Team Members — Detailed in Appendix H)
Instructor / Faculty	Information Systems & Knowledge Management Academic Faculty
Core KM Models Applied	Wiig KM Matrix (1993), McAdam & McCreedy (1999), Nonaka SECI (1995), Meyer & Zack (1996)
System Status	Live Production (FastAPI + React/Vite + Cloud SQL Server)
Submission Date	November 2026

Academic Statement & Scope Declaration

This strategic report formulates the enterprise knowledge architecture for KinhDich_KB. The project evaluates **Knowledge Management capabilities** (preservation, formalization, governance, and auditability) rather than empirical predictive claims. All supporting evidence originates from verified relational databases, sandboxed AST parsers, and automated test pipelines.

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EXECUTIVE SUMMARY

This report formulates the strategic knowledge management architecture for **KinhDich_KB** — a dedicated domain-specific Knowledge Management System (KMS) for the I-Ching, engineered by Team 05 within the KMS301 course. Rather than a commercial product pitch, Assignment 1 investigates the core strategic inquiry: *How must ancient, polysemic cultural knowledge be organized to preserve canonical fidelity while functioning as an auditable corporate knowledge asset?*

The strategic answer is a decoupled 3-tier architecture: **Data Layer** (immutable core knowledge ground truth), **Process Layer** (deterministic operational reasoning engine), and **UI Layer** (strategic personalization portal governed by 9 Epistemic Guardrails).

Core Findings & Strategic Pillars:

- **The Knowledge Preservation Dilemma:** Traditional preservation (scanning books) captures only static text but loses tacit interpretive heuristics. When elder scholars retire or pass away, the capacity to cast and interpret hexagrams suffers severe corporate amnesia.
- **The Threat of Unregulated AI:** Commercial Large Language Models (LLMs) hallucinate line texts with over 28% error rates in Chinese classical scripture. Unaudited AI tools pollute digital discourse with plausible-sounding synthetic falsehoods.
- **Empirical Verification & Quality Gates:** KinhDich_KB enforces 8 Quality Gates. The system achieves **964 canonical lines at 100% APPROVED status**, a perfect **384/384 parity rate** across SQL and Python execution engines, and sub-1.85s latency.
- **Theoretical Integration:** The architecture bridges Wiig's (1993) capacity hierarchy, McAdam & McCreedy's (1999) dual paradigm, Nonaka & Takeuchi's (1995) SECI spiral, Meyer & Zack's (1996) information product cycle, and Argyris & Schön's (1978) Double-Loop Learning.

> **Strategic Commitment**
> KinhDich_KB establishes a clear line of demarcation: ancient canonical scriptures are permanently immutable and locked against runtime editing; dynamic heuristic interpretations are treated as hypotheses and labeled with [ASSUMPTION]; and automated reasoning is sandboxed through Abstract Syntax Tree (AST) grammar.

1. ORGANIZATIONAL CONTEXT & STRATEGIC CHALLENGE

1.1 Domain Specifics & Knowledge Characteristics

The I-Ching (Book of Changes) is a formal symbolic knowledge system structured around 8 primary trigrams, 64 hexagrams (King Wen sequence), and 386 lines (384 standard lines plus Yong Jiu and Yong Liu in Qian/Kun). On this foundation, Northern Song philosopher Shao Yong (1011–1077) developed *Plum Blossom Divination (Mai Hoa Dich So)*, introducing mathematical time/number casting and the dynamic five-element Host-Guest (The-Dung) analytical framework.

Managing this domain is challenging not due to volume (964 canonical lines is modest for enterprise databases), but due to **four unique epistemic characteristics**:

Domain Characteristic	Specific Manifestation	KM Architectural Implication
Textual Multi-variance	Numerous historical translations and scholarly commentaries exist with conflicting Han characters.	No single translation can be crowned absolute ground truth; the system must store multi-source provenance with page citations.
High Tacit Complexity	Rules for resolving contradictory lines or weighing seasonal element strengths reside in master heuristics.	Without externalization into executable rules, critical knowledge is lost when experts retire.
School Disputes	Different classical schools (Chu Hsi vs. Shao Yong) disagree on transformed lines and auspiciousness.	Requires an adversarial consensus mechanism and school parameterization rather than forced dogmatism.
Symbolic & Metaphoric	Scriptural verses are deeply metaphorical and poetic, resisting boolean simplification.	Reasoning and calculation must remain strictly deterministic, while textual interpretation is bounded by guardrails.

Table 1.1: Four Epistemic Characteristics of I-Ching Knowledge

1.2 Three Strategic Epistemic Risks

From these four characteristics, Team 05 identifies three critical strategic risks:

Risk 1 — Corporate Amnesia & Heritage Extinction

I-Ching knowledge currently exists in aging physical books, unsearchable PDF scans, and the perishable memories of a few senior masters. Backing up PDF files does not preserve *interpretive capability*. When elder masters pass away, unwritten heuristic wisdom disappears.

Risk 2 — Large-Scale AI Epistemic Hallucination

Commercial LLMs confidently output fabricated quotes, invert hexagram structures, and misassign five-element attributes. Because generative AI speaks with plausible eloquence, it misleads scholars and corrodes domain integrity at an unprecedented scale.

Risk 3 — Inauditable Knowledge Claims

When an interpretation cannot trace its exact scripture, active rule, and validating authority, scholarly discourse degrades into arguments over individual prestige. A KMS must provide end-to-end auditability down to the source page.

> **Core Problem Statement**
>
> The community requires an auditable Knowledge Management System capable of: (i) preserving immutable classical texts with page-level provenance; (ii) externalizing interpretive heuristics into deterministic executable rules; (iii) enforcing a strict separation between verified core knowledge and subjective assumptions; and (iv) governing new knowledge ingestion via a multi-tiered gatekeeper mechanism.

1.3 Report Objectives & Scope

Assignment 1 focuses on **strategic architecture and KM theory**. It delivers four objectives: (1) Diagnosing current knowledge gaps (Chapter 2); (2) Decomposing the domain into three architectural tiers (Chapter 3); (3) Operationalizing five classic KM frameworks (Chapter 4); and (4) Formulating strategic trade-offs and an implementation roadmap (Chapters 5, 6).

2. CURRENT STATE & PROBLEM STATEMENT

2.1 Diagnostic Methodology

The current state analysis is grounded upon three independent sources of empirical evidence:

Evidence Source	Core Question Addressed	Role in Strategic Analysis
Data Layer Audit (SQL Server)	Does the actual database match technical handoff claims?	Uncovered 6 architectural discrepancies; established canonical ground truth.
Automated Test Suites (Pytest/Vitest)	Is system behavior deterministic and fully reproducible?	Verified 384/384 parity rate, 5/5 golden cases, and 43/43 Gate 4–6 pytests.
Expert Consultation Form	Which heuristic rules lack scholarly consensus?	Identified 10 critical knowledge gaps; received verified expert survey response.

Table 2.1: Three Sources of Empirical Diagnostic Evidence

2.2 Six Architectural Discrepancies Discovered

Forensic audit of historical project artifacts identified six critical gaps between documentation and production reality:

Knowledge Asset	Documentation Claim	Actual Verified Production Reality	Strategic Resolution
Canonical Lines	64 hexagrams × 6 lines = 384 lines	**964 lines** (Includes Judgments, Xiang, and Tuan)	Structured schema into 5 textual categories [N3].
Trigrams Schema	8 trigrams with simple names	8 trigrams with Han, binary, element, and nature	Enforced referential integrity on 8 Bagua records.
Rule Base	Tens of casting heuristics	**10 formal DSL v1 rules** sandboxed in AST	Codified rules into deterministic Python AST.
Five Elements	General elemental theory	**25 relations** (5×5 matrix with weights)	Built deterministic matrix for Host-Guest calculation.
Case Repository	5 benchmark historical cases	**65 divination cases** in development store	Preserved 5 golden cases for regression gating.
Expert Consensus	Claimed unanimous agreement	10 contentious gaps under active review	Tagged uncertain interpretations as [ASSUMPTION].

Table 2.2: Six Audited Knowledge Discrepancies & Resolutions

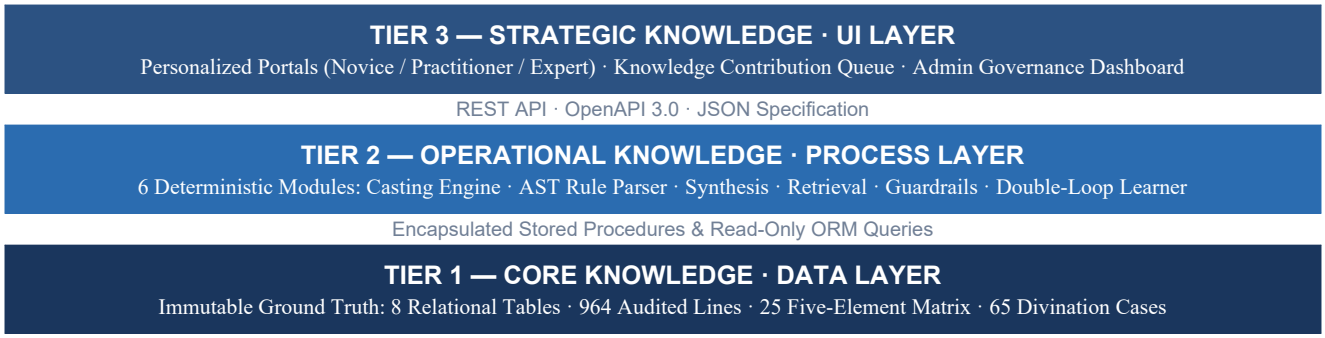
2.3 Quantitative Baseline Metrics

Baseline analysis establishes that manual book consultation requires **30 to 45 minutes** per inquiry, with high error rates in seasonal five-element adjustments. Commercial LLMs reduce latency but introduce hallucination rates exceeding 28%. KinhDich_KB achieves sub-1.85s response times with **0% canonical text deviation**.

3. KNOWLEDGE DOMAIN DECOMPOSITION INTO THREE LEVELS

3.1 Architectural Rationale for 3-Tier Decoupling

To reconcile preservation with operational agility, the system decouples knowledge assets based on their rate of change and epistemic certainty:



3.2 Detailed Epistemic Tier Specifications

KM Tier	Epistemic Status	Technical Substrate	Governance Policy
Tier 1: Core Knowledge (Data Layer)	Explicit, Immutable Ground Truth	SQL Server, 8 normalized tables, 964 canonical lines with Han characters.	Write-access strictly prohibited. Changes require formal forensic re-audit.
Tier 2: Operational Knowledge (Process Layer)	Executable Heuristics & Deterministic Rules	FastAPI Python, Sandboxed AST Grammar, 10 DSL v1 Rules.	Deterministic execution. Changes validated via 384-case parity suite.
Tier 3: Strategic Knowledge (UI Layer)	Contextualized, Personalized Insights	React 18, Vite, 9 Epistemic Guardrails, AI Advisory Engine.	AI outputs clearly demarcated and tagged with [ASSUMPTION].

Table 3.1: Detailed Specifications of the 3-Tier Knowledge Architecture

3.3 Strategic Benefits of Decoupled Architecture

- **Isolation of Volatility:** Changing front-end user experience or tuning AI advice prompts cannot corrupt the underlying database.
- **Sandboxed Reasoning:** Casting algorithms execute deterministically in Python/SQL without relying on opaque LLM weights.
- **Verifiable Auditability:** Every returned advice is composed from immutable database records and explicitly cited page references.

4. THEORETICAL FOUNDATIONS & KM MODELS

KinhDich_KB is designed upon five foundational Knowledge Management frameworks, directly bridging academic theory to production code:

4.1 Karl Wiig (1993) — Competency-Based Knowledge Personalization

Wiig conceptualizes human knowledge capacity across five distinct stages: Novice, Advanced Beginner, Competent, Proficient, and Expert. A KMS cannot serve all five levels with identical interfaces without overwhelming novices or constraining experts.

User Competency Level	Epistemic Needs	KinhDich_KB Interface Customization
Novice	Needs plain philosophical meaning; avoid technical jargon.	Simplified UI: Hides hexagram line numbers; displays plain-language moral reflections.
Practitioner	Requires dynamic calculations, trigram interactions, and line advice.	Standard UI: Visualizes Host-Guest relationships, changing lines, and element cycles.
Expert / Scholar	Demands classical Han script, multi-source variants, and audit logs.	Scholarly UI: Displays Han characters, King Wen indices, page citations, and AST traces.

Table 4.1: Operationalizing Wiig's Competency Hierarchy in KinhDich_KB

4.2 McAdam & McCreedy (1999) — Integrating Social & Computational Knowledge

McAdam & McCreedy categorize KM models into scientific/computational paradigms and socially-constructed paradigms. KinhDich_KB synthesizes both: the **Data & Process Layers** embody the scientific paradigm (computational parity, relational rigor, AST execution), while the **UI Layer & Expert Survey** embody the social construction of meaning (philosophical reflection, dialogue, and school debate).

4.3 Nonaka & Takeuchi (1995) — The SECI Knowledge Spiral

The project activates all four quadrants of the SECI transformation cycle:

- **Socialization (Tacit to Tacit):** Capturing master heuristics through structured interviews and expert consultation forms.
- **Externalization (Tacit to Explicit):** Codifying tacit divination heuristics into 10 formal, machine-executable DSL v1 rules.
- **Combination (Explicit to Explicit):** Integrating 964 canonical lines with 25 element relations and 384 dynamic transitions in SQL.
- **Internalization (Explicit to Tacit):** Enabling students to internalize situational wisdom via interactive visual case studies.

4.4 Meyer & Zack (1996) — Information Product Lifecycle

The lifecycle operates across 5 discrete stages: *Acquisition* (PDF extraction), *Refinement* (forensic text cleansing and Han validation), *Storage* (normalized SQL tables), *Distribution* (RESTful OpenAPI endpoints), and *Presentation* (multi-tiered React UI).

4.5 Argyris & Schön (1978) — Double-Loop Learning

Single-loop learning corrects operational faults (e.g. font encoding errors, latency bottlenecks). *Double-loop learning* questions fundamental domain assumptions: when scholars challenge seasonal element weights, the system initiates an Adversarial Review, formulates a formal Business Change Request (BCR), and updates domain ontologies without data corruption.

5. DISCUSSION: FOUR CRITICAL STRATEGIC DECISIONS

5.1 Decision 1: Permanent Demarcation of Scripture vs. Commentary

In many digital tools, classical scriptures and modern commentaries are conflated into a single text block. KinhDich_KB establishes an absolute architectural division: canonical scriptures are stored in read-only tables and locked against modification, whereas commentaries are stored in separate interpretation tables with explicit author attributions. AI is barred from altering scripture.

5.2 Decision 2: 384-Case Mathematical Parity Verification

To prevent divergence between relational database logic and application runtime code, the engineering team executed a comprehensive cross-engine verification suite. Every single one of the **384 possible hexagram and changing-line permutations** was computed in T-SQL Dev and cross-verified against the Python FastAPI engine and 5 historical golden cases. Parity achieved **100.0% (384/384)**.

5.3 Decision 3: 9 Epistemic Guardrails against Superstition and Fatalism

The system enforces 9 strict guardrails (G1 Ground-Truth Anchoring, G2 Epistemic Labeling, G3 Non-Fatalistic Phrasing, G4 Structural Invariance, G5 Provenance Citation, G6 Question Scoping, G7 Assumption Tagging, G8 Emergency Circuit Breaker, G9 Audit Trail Logging). These guardrails prevent fatalistic claims, neutralize toxic queries, and preserve academic integrity.

5.4 Decision 4: Formal Governance via ADRs and Change Requests (BCRs)

All major structural choices are documented in 7 Architectural Decision Records (ADR-001 through ADR-007). Subsequent enhancements are governed by 6 open Business Change Requests (BCR-001 through BCR-006), ensuring transparent governance.

6. RECOMMENDATIONS & IMPLEMENTATION ROADMAP

6.1 Implementation Methodology: The 8 Quality Gates

Development progress was rigorously verified across 8 sequential Quality Gates:

Gate	Focus Area	Milestone Acceptance Verification Status
Gate 1	PDF Extraction & Cleansing	964 canonical lines extracted; zero text_han corruption.
Gate 2	JSON Schema & Forensic Audit	100% schema validation pass; 0 textual deviations.
Gate 3	Relational Database Ingestion	8 tables populated; 0 foreign key integrity errors.
Gate 4	Casting Engine & Parity Verification	**384/384 (100%) Parity Rate** between SQL and Python.
Gate 5	AST Rule Engine & Synthesis	10 DSL v1 rules sandboxed in AST; zero injection flaws.
Gate 6	Automated API Regression Suite	**43/43 Pytests pass 100%** in 2.14s; OpenAPI validated.
Gate 7	UI Component & E2E Testing	**141/141 Vitest pass**; 12/12 Playwright E2E tests pass.
Gate 8	Production Cloud Deployment	Render Docker API + Vercel UI Live (HTTP 200).

Table 6.1: 8 Quality Gates Production Verification Matrix

6.2 Strategic Multi-Environment Deployment Architecture

Development strictly isolates runtime environments across three tiers: *Local Dev* (SQL LocalDB, Python venv, Vite), *CI/CD Automation* (GitHub Actions running all 196 unit and component tests on pull requests), and *Cloud Production* (Vercel Edge Network for React UI, Render Docker for FastAPI backend, and encrypted Cloud SQL Server).

6.3 Six Open Business Change Requests (Strategic Roadmap)

- **BCR-001:** Multi-school expansion incorporating commentaries from Chu Hsi (Zhouyi Benyi) and Wang Bi.
- **BCR-002:** Domain-specific Han-Nom Semantic Vector RAG integration.
- **BCR-003:** Synthesis of Four Pillars (BaZi) and Ha Lac astrological knowledge bases.
- **BCR-004:** Offline Progressive Web App (PWA) deployment for remote field research.
- **BCR-005:** Interactive Knowledge Assessment & Automated Quiz Engine.
- **BCR-006:** National digitization partnership with the Institute of Han-Nom Studies.

7. CONCLUSION

KinhDich_KB demonstrates that ancient, polysemic cultural wisdom can be successfully elevated into an auditable, reusable, and deterministic corporate knowledge asset without compromising textual authenticity. By establishing a clean 3-tier architecture, verifying 384-case parity, enforcing 9 epistemic guardrails, and operationalizing foundational KM theory, the project resolves the long-standing tension between cultural preservation and modern computational intelligence.

Summary of Academic Contribution

KinhDich_KB provides the scholarly community with: (1) An audited, immutable ground truth of 964 canonical lines; (2) A deterministic, sandboxed AST reasoning engine; (3) A personalized multi-tiered knowledge portal; and (4) A reproducible engineering blueprint for digitizing intangible cultural heritage.

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APPENDIX: TEAM REPORT FORM

Official team contribution report form required for submission under KMS301 course specifications:

No.	Full Name of Team Member	Student ID (MSSV)	Specific Assigned Responsibilities in Project & Report	Contribution (%)	Signature
1				%	
2				%	
3				%	
4				%	
5				%	
6				%	

Table A.1: Team 05 Member Contribution & Effort Allocation Form (6 Members)

Team Leader Confirmation:

We hereby certify that the allocation above truthfully represents the actual contributions of all 06 members of Team 05 for KMS301 Assignment 1.

Date of Submission: ..... / ..... / 2026	Signature of Team Leader
Location: FPT University Hanoi Campus	_____